"Relative Histories" Formulation of Quantum Mechanics III: Further Mathematical Development

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The relative histories formulation (RHF) is a novel formulation of quantum mechanics based upon the assumption that the physical state of the universe can be represented mathematically by an ensemble E of four-dimensional manifolds $\mathcal{W}$. In a companion paper (Part I [27]), the basic formalism of the RHF is presented and demonstrated explicitly to make the same predictions as any of the other standard formulations of quantum mechanics. In a second companion paper (Part II [28]), the interpretational issues surrounding the RHF are explored. In the present paper (Part III), the mathematical underpinnings of the RHF are developed beyond the presentation of Parts I and II. We present a definition of the probability measure of an ensemble $E^W_{\vec{\alpha}}$ and show that this definition is equivalent to the random walk hypothesis presented in Part I. The connection between the decoherence functional of the consistent histories formulation and the probability measure over $E^W_{\vec{\alpha}}$ is discussed. $E^W_{\vec{\alpha}}$ are shown to display the self-similar properties of fractals. In addition, we show that the elements of $\mathcal{O}$-state space, $\vec{\alpha}$, are infinite dimensional and form a Hilbert space. The properties of the fundamental group of the four-manifolds $\mathcal{W}_{\vec{\gamma}}$ are explored as we work towards a compact definition of these manifolds. Connections to string theory, derivation of the Second Law of Thermodynamics, the arrow of time, and CPT violation are also discussed.

PACS numbers: PACS numbers go here. CHF, consistent histories formulation; RHF, relative histories formulation; QM, quantum mechanics; GR, general relativity; FPI, Feynman path integral; MWI, multiple worlds interpretation; EPR, Einstein-Podolsky-Rosen; dBB, de Broglie-Bohm; CPC, causality preservation conjecture; ER, ensemble-reality interpretation of the RHF; MR, manifold-reality interpretation of the RHF

I. INTRODUCTION

Introduction.

II. PROPERTIES OF THE VARIOUS SPACES OF THE RHF

In this section, we explore in greater detail the properties of the various spaces and other mathematical entities of the RHF that were introduced in the preceeding section. We remind the reader that these mathematical entities, and the interrelations between them, are summarized in Fig. 6.

A. Properties of 4-manifolds $\mathcal{W}$

1. *Each element of $E^{\mathcal{W}}_{FUND}$ has a distinct topology*

Using the method of analytic continuation [31] that was discussed previously, the value of $g_{ab}(0,0,0,0)$ and its derivatives can be used to define the function g_{ab} throughout the *entire* manifold $\mathcal{W}$, where $\mathcal{W}$ is any manifold that contains a point with that power series. In other words, given any two $\mathcal{W}_{\vec{\gamma}'}$, $\mathcal{W}_{\vec{\gamma}''} \in E^{\mathcal{W}}_o$, the method of analytic continuation can be used to show that $\mathcal{W}_{\vec{\gamma}'}$ is everywhere equal to $\mathcal{W}_{\vec{\gamma}''}$. However, this line of reasoning is based upon the assumption that $\mathcal{W}_{\vec{\gamma}'}$ and $\mathcal{W}_{\vec{\gamma}''}$ have the same topology. Therefore, for any two $\mathcal{W}_{\vec{\gamma}'}$, $\mathcal{W}_{\vec{\gamma}''} \in E^{\mathcal{W}_o}$ where $\mathcal{W}_{\vec{\gamma}'}$ is not equal to $\mathcal{W}_{\vec{\gamma}''}$, we conclude that $\mathcal{W}_{\vec{\gamma}'}$ and $\mathcal{W}_{\vec{\gamma}''}$ must have *different topologies*.

2. *Classical vacuum energy density*

We note in passing that it is attractive to speculate that the lagrangian at a point in $\mathcal{W}_{approx}$ (the classical lagrangian) may be related in some fashion to the "density" of the quantum foam in the vicinity of the corresponding point in $\mathcal{W}_{\vec{\gamma}}$ – averaged, perhaps, over each $\mathcal{W}_{\vec{\gamma}}$ in $E^{\mathcal{W}}$. This has been addressed, for example, in [18]. To pursue this line of thought, it will be necessary to specify in greater detail than is done in this paper the topological properties of the world manifolds $\mathcal{W}_{\vec{\gamma}}$ that populate $E^{\mathcal{W}}_{FUND}$. Therefore we will not explore this topic further in this paper.

B. Properties of $\mathcal{O}_X$, $E^{\mathcal{O}}_X$, $\mathcal{O}_o$, and $E^{\mathcal{O}}_{FUND}$

1. *The number of elements in $E^{\mathcal{O}}_X$ is some high order of infinity*

Recall that we will often use the symbol $\mathcal{O}_X$ to represent the state of a macroscopic object, such as a macroscopic experimental apparatus. For example, for the analysis of a statement X = "The laboratory is set up and ready to measure the spin state of a particle," $\mathcal{O}_X$ would represent the entire laboratory, including a particle source and a Stern-Gerlach apparatus. Recall also that $\mathcal{O}_X$ provides a specification of the exact state of this macroscopic object, down to the tiniest detail, including the topology of the spacetime foam. In general, then, we will require a very large number of objects $\mathcal{O}_X$ to populate the ensemble $E^{\mathcal{O}}_X$. Each of these different $\mathcal{O}_X$ will differ in the small details, such as the exact momenta and positions of

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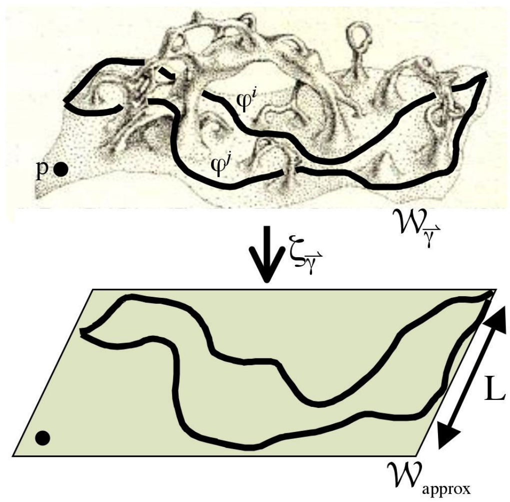

FIG. 1: Top: A two dimensional hyperslice through a four dimensional manifold $\mathcal{W}_{\vec{\gamma}}$ illustrating non simple-connectivity, which has been proposed as a model of quantum foam. We assume that these fluctuations in the curvature of spacetime exist (mostly) on a scale below the Planck-Wheeler length $L = \sqrt{\frac{G\hbar}{c^3}}$ [10]. (Adapted from [30], Fig. 14.3 (c).) Bottom: For any such manifold $\mathcal{W}_{\vec{\gamma}}$ there exists a mapping $\zeta_{\vec{\gamma}}$ that takes points p or curves φ in $\mathcal{W}_{\vec{\gamma}}$ onto an approximately flat, simply connected manifold $\mathcal{W}_{approx}$. ($\mathcal{W}_{approx}$ may be referred to as the "universal covering manifold;" see, for example, [11].) In effect, the small scale fluctuations of $\mathcal{W}_{\vec{\gamma}}$ are "smoothed out" by the mapping to $\mathcal{W}_{approx}$.

individual particles. They will typically even be of *differing topologies* (and will virtually *all* be multiply connected, owing to the nature of spacetime foam). However, they will by definition agree in the essentials. The only requirement is adherence to the wording of the statement X. (We may provide as much detail as we desire to the statement X.) Since there are obviously many ways to vary the appearance of the small details – the spacetime foam, the exact positions and momenta of individual particles – while still adhering to the statement X, we see that there will typically be a very large number of distinct objects $\mathcal{O}_X$ that populate $E^{\mathcal{O}}_X$. In fact, this may very well be some large order of infinity. Later we will see that our world ensembles, as well, will likely be composed of some high order of infinite number of elements. In the case of the world ensembles, this will turn out to be a useful trait, as we will see.

If this seems to present a conceptual difficulty to the reader, we point out that part of the difficulty arises due to the fact that we are accustomed to defining quantum mechanical experiments in terms of clicks and dials on macroscopic measuring devices. In the RHF, there is no reason that we cannot continue in this same vein, that is, to define the observer as a macroscopic device such as a computer, and to set up an experiment via construction of a statement X that also makes reference to macroscopic objects like Stern-Gerlach apparati. However, the RHF makes no requirement that any of these relevant entities be either macroscopic or microscopic. As we will see, some of the mathematics will become simplfied and more tractable if we shrink some of these entities down to the microscopic realm. So for conceptual purposes and for the sake of making the concepts of the RHF easier to accept from a psychological standpoint, we will continue to make use of macroscopic objects in our deliberations, but will alternatively think in terms of microscopic objects in cases where mathematically appropriate.

2. *A compact definition of the observer*

In Sec. **??**, we stipulate that in theory, *any* $\mathcal{O}_{\vec{\alpha}}$ – that is, *any* element of $E^{\mathcal{O}}_{FUND}$ – can play the role of "observer." However, in this section, we will present some restrictions on the mathematical specification of the observer – in particular, we restrict the topology of the observer to be simply connected. The purpose of this restriction, we will see, is that it will allow us to refine our definition of the state space of the observer. In particular, we will be able to equate our observer with a point in an infinite dimensional state space. (Alternatively, with a line segment of arbitrarily small length through a ten dimensional state space.)

Our generalized mathematical specification for an observer $\mathcal{O}_o$ will begin as follows. Consider a point $p = (0,0,0,0)$ in a four-dimensional spacetime, located at coordinates $(0,0,0,0)$ with metric $g_{ab}(0,0,0,0)$. Suppose we are additionally provided with the entire *power series* of g_{ab} at p (that is, all of the derivatives and partial derivatives, up to infinite order, of $g_{ab}(0,0,0,0)$.) Consider a four dimensional ball of radius r centered around the point p. This four-ball is of course a 4-object in the language of the RHF, so we will label it as such: $\mathcal{Q}_{p,r}$, with p the center of the 4-ball and r its radius. Recall that we assume that the metric $g_{ab}(x_1, x_2, x_3, x_4)$ is everywhere analytic and nowhere singular. Using Taylor's Theorem (see the Appendix, Sec. A), we can express $g_{ab}(x_1, x_2, x_3, x_4)$ at *any point* in $\mathcal{Q}_{p,r}$ as a function of the power series of g_{ab} at p. Of course, this is only possible if we restrict the topology of the 4-ball to have some particular topology. So for simplicity, we will restrict it to be simply connected.

Recall that our manifolds are not necessarily time-orientable, which means that there is no particular direction that can be assigned the time direction throughout the manifold. $\mathcal{Q}_{p,r}$, however, is time orientable (since it is simply connected). Thus, let us arbitrarily assign one direction $\vec{t}$ as corresponding to the "time" axis. $\vec{t}$ is a four-vector, which we will refer to as the "temporal support" of $\mathcal{Q}_{p,r}$. We will use the scalar τ to refer to the location along the $\vec{t}$ axis. (This is the same τ that was discussed in Sec. **??** and Fig. **??**.) Given this specification, we can define the hyperslice through $\mathcal{Q}_{p,r}$ at the coordinate $\tau = 0$, which gives us a 3-ball with radius r centered at the point $(0,0,0)$ which we label as a 3-object, $\mathcal{O}_{p,r}$.

Our proposal, then, is to require that our observer $\mathcal{O}_o$ be able to be cast in the form: $\mathcal{O}_{p,r}$. As we have discussed previously, a macroscopic object will typically be multiply connected, owing to the multiple connectedness of the background spacetime foam. To adhere to the notion that $\mathcal{O}_{p,r}$ is simply connected, therefore, we will envision our $\mathcal{O}_{p,r}$'s as being very (in fact, arbitrarily) small. With this viewpoint in mind, any $\mathcal{Q}$ that gives us one possible time-dependent evolution of $\mathcal{O}_o = \mathcal{O}_{p,r}$ (as depicted in our tree diagrams) will be envisioned as being in the shape of "long thin tubes."

3. *Temporal support, $\vec{t}$*

Let us reiterate the notion of temporal support. It is possible to have two objects, $\mathcal{O}_1$ and $\mathcal{O}_2$, that exist at the same place in $\mathcal{O}$-State Space and yet have different temporal support. In other words, their respective power series may be identical; yet they may have different temporal support. For example, we might specify that the time dimension for $\mathcal{O}_1$ corresponds to x_4 i.e.: $\vec{t_1} = (0, 0, 0, 1)$, while the time dimension for $\mathcal{O}_2$ corresponds to, say, a linear combination of x_1 and x_3, e.g.: $\vec{t_2} = (\sqrt{\frac{1}{2}}, 0, \sqrt{\frac{1}{2}}, 0)$. (Note that we have defined $\vec{t}$ as a *unit* vector so that $|\vec{t}| = 1$.) Thus, the complete specification of the state of an object must include not only its location in state space, but also its temporal support.

4. *$\mathcal{O}$-State Space*

According to the above considerations, the state of $\mathcal{O}_o$ can be fully specified simply by providing $g_{ab}(0, 0, 0, 0)$ and its derivatives (as well as its temporal support $\vec{t}$). This makes $\vec{\alpha}$ an infinite dimensional vector, since the number of derivatives of g_{ab} is infinite.

To show this explicitly, let us write out explicitly the components of $\vec{\alpha}$. g_{ab} is a 4×4 matrix with sixteen components. However, the metric tensor g_{ab} is, by definition, symmetric (that is, $g_{ab} = g_{ba}$) and therefore actually requires only ten components rather than sixteen for its complete specification. Denote these ten components using the notation $g_j(p), j = 1, 2, 3, \ldots, 10$. To perform a Taylor series expansion about the point p, we must specify the value of each of these ten components as well as all combinations of partial derivatives. That is, we must specify: $Del^i g_j(p), j = 1, 2, \ldots, 10; i = 0, 1, 2, \ldots, \infty$

C. Properties of the tree diagram

1. *Merging, fusing, and looping*

We will attempt to minimize the restrictions that we place on the potential makeup of tree diagrams. For example: we do not forbid merging (or "fusing") (as opposed to "splitting"), as depicted in Fig. 2 (for $\vec{\alpha}'$). We also do not forbid the possibility that the time-dependent evolution of the state of an object may take it to a state indistinguishable from a prior state – in other words, it may loop back upon itself to arrive again at a prior point in state space, as depicted in Fig. 2 (for $\vec{\alpha}''$).

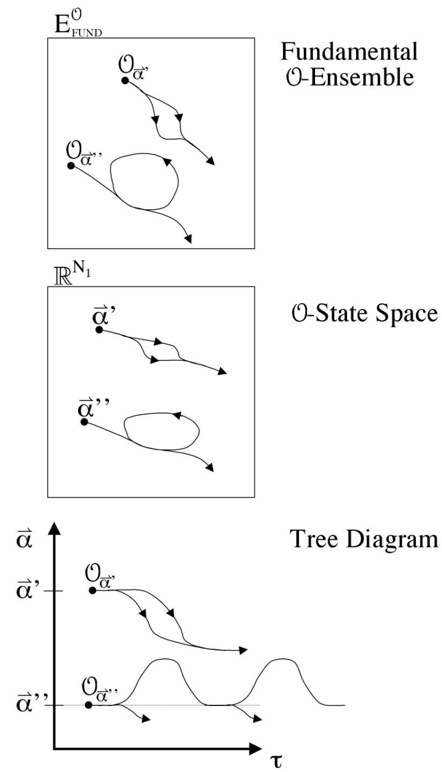

FIG. 2: $\mathcal{O}_{\vec{\alpha}'}$: Ability of two paths through state space to "merge" or "fuse" into the same state. $\mathcal{O}_{\vec{\alpha}''}$: Ability of the state of a system to "loop around" to return to a former state.

2. *For every point $\vec{\alpha}$ in state space, there exists a trajectory back to itself*

In QM, strictly speaking, the wavefunction of any given particle in spatial coordinates is never exactly zero – it can, at most, merely approach zero. Thus, there is always a nonzero probability that a particle might tunnel to, basically, anywhere. This is true for every particle in the universe. Therefore, given *any* two arbitrary states of the universe, Ψ_1 and Ψ_2, there is always a nonzero (even if *vanishingly small*) chance that the universe (assuming it to be in the state Ψ_1) could evolve to the state Ψ_2.

In the language of the RHF, the state of the universe is specified via the ensemble $E^{\mathcal{W}}_{\vec{\alpha}}$, which is defined in terms of our observer, $\mathcal{O}_{\vec{\alpha}}$, or equivalently, a location $\vec{\alpha}$ in state space. If we are to remain consistent with the preceeding paragraph, then we conclude that: Given any two locations $\vec{\alpha}'$ and $\vec{\alpha}''$ in RHF state space, there must always exist at least one path through state space from $\vec{\alpha}'$ to $\vec{\alpha}''$. Since this is also true vice versa – that is, there must always exist at least one path through state space from $\vec{\alpha}''$ to $\vec{\alpha}'$ – then there must always be

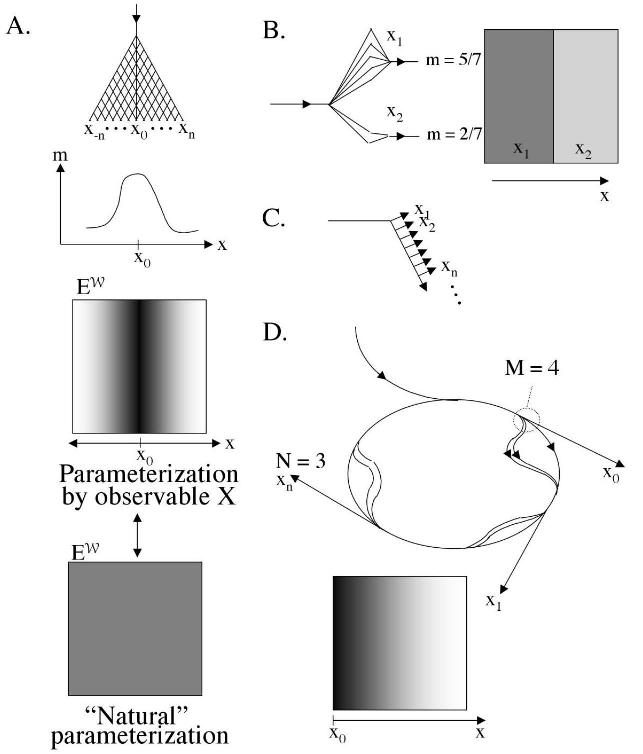

FIG. 3: Different tree diagram configurations can be seen to correspond to different probability distribution functions. Depicted are tree diagram configurations that give rise to (A) gaussian, (B) proportionality equation, (C) delta function, and (D) exponential.

at least one path through state space from $\vec{\alpha}'$ back to itself, $\vec{\alpha}'$. Therefore, for every point $\vec{\alpha}$ in state space, there exists at least one trajectory back to itself, which we call a "loop."

3. *Tree diagrams and probability distribution functions*

Let us consider some various configurations that the tree diagram may assume, and examine how these configurations influence the probability of collapse onto an observable X. These configurations are depicted in Fig. 3. In each of these tree diagrams, we may imagine that the observable X corresponds to the observed location of a particle on a detector apparatus. As discussed previously, we will present the tree diagram and the observable X as being discrete valued, but may also convert X into a continuous variable by allowing the branch structure of the tree diagram to approach infinity in an appropriate manner that preserves its essential characteristics. (How exactly we do so may differ from tree diagram to tree diagram, and should be obvious from the context for the simple examples presented here.) We will find that we are able to devise simple tree structures to correspond to a surprisingly rich variety of functions over the probability distribution. The primary purpose of this exercise, therefore, is to develop an appreciation for the versatility of the RHF approach. In addition, this exercise will help us in basic understanding of the concepts introduced in the preceeding sections, such as statements, observables, and relative probability.

In Fig. 3, we present several different potential configurations of the tree diagram along with their associated ensembles, which we have parameterized along one axis (where indicated) using the value of the (sometimes continuous) observable X. The shading of the ensembles represents the relative probability measure of collapse onto X. Through application of the random walk hypothesis to the tree diagram of Fig. 3 A, for example, it should be evident that collapse onto values of x close to x_0 are more likely than collapse onto values far from x_0. This is depicted in the shading of the relative world ensemble as well as in the graph of the probability measure as a function of x, which takes the form of a gaussian distribution: $m(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$ (Fig. 3 A). Other tree diagram formations may be shown to give rise to other probability distribution functions: a proportionality equation $m(x_i) = a_i x_i$ (Fig. 3 B), a delta function $m(x) = \delta(x)$ (Fig. 3 C), or an exponential $m(x) = e^{-kx}$ (Fig. 3 D). (We leave it as an exercise to the reader to demonstrate that the tree diagram of Fig. 3 D does in fact give rise to an exponential. If N equals the number of branches that leave the loop, and M equals the number of branches at each of the sub branch points in the diagram, and M and N are related by the equation $M = kN$, then if we take the limit as N approaches infinity while keeping k constant, then k is the decay rate of the corresponding exponential: e^{-kx}.)

4. *Minimization laws*

We wish to make a special point about the delta function in Fig. 3 C. If the value of the variable x_i increases for increasing i, then the tree diagram in Fig. 3 C will make small values of x more probable than larger values of x; that is, it will tend to *minimize* the variable x. If the value of x_i *decreases* for increasing i, then we will, conversely, *maximize* x. If we imagine that x represents the classical action S, then we see how Hamilton's Principle of Least Action may arise from the RHF. In fact, this may be used to understand the origin of *any* of the extremal laws. Keep in mind that according to this scheme, it is *possible* to realize *any* value of our observable x. When we say that the tree diagram minimizes or maximizes x, we are speaking in probabilistic terms.

D. Properties of $E^{\mathcal{W}}_{FUND}$ and $E^{\mathcal{W}}_{\vec{\alpha}}$

1. *For any given $\mathcal{W}_{\vec{\gamma}}$ and $E^{\mathcal{W}}_{\vec{\alpha}}$, $\mathcal{W}_{\vec{\gamma}}$ is an element of $E^{\mathcal{W}}_{\vec{\alpha}}$*

Recall our notion (depicted in Fig. **??**) that collapse always proceeds from an ensemble to a subsensemble. Given that there must always exist at least one path through state space from any $\vec{\alpha}'$ to any other $\vec{\alpha}''$ (as discussed in Sec. II C 2), if $\mathcal{W}_{\vec{\gamma}}$ is any arbitrary element of $E^{\mathcal{W}}_{\vec{\alpha}''}$, then $\mathcal{W}_{\vec{\gamma}}$ must also be an element of $E^{\mathcal{W}}_{\vec{\alpha}'}$. Therefore, if $\mathcal{W}_{\vec{\gamma}}$ is an element of *any* $E^{\mathcal{W}}_{\vec{\alpha}}$,

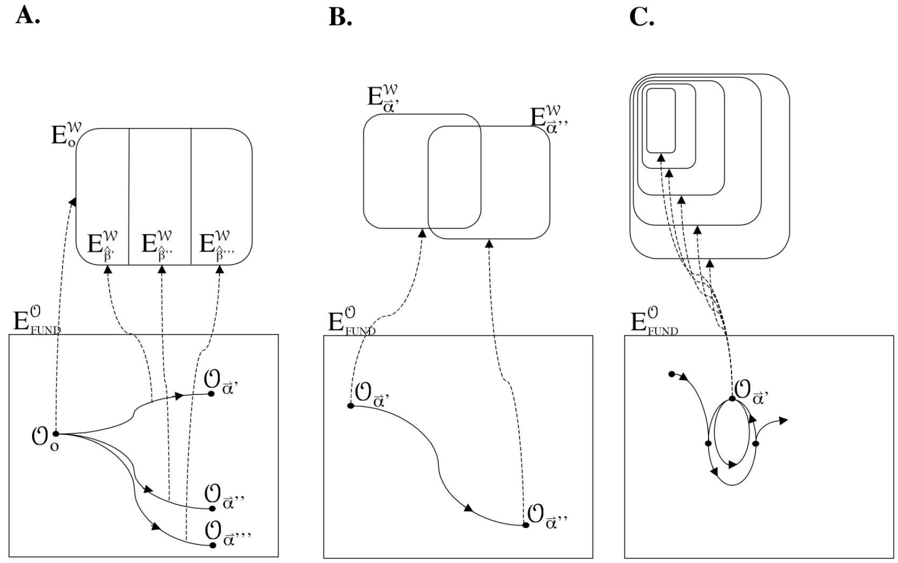

FIG. 4: Bottom: Various examples of different types of trajectories through state space. Top: World ensembles relative to some of the three-objects $\mathcal{O}$ or four-objects $\mathcal{Q}$ of these trajectories.

then it must be an element of *every* $E^{\mathcal{W}}_{\vec{\alpha}}$. Therefore, given any two $E^{\mathcal{W}}_{\vec{\alpha}'}$, $E^{\mathcal{W}}_{\vec{\alpha}''}$, the difference between them cannot be *which* world manifolds are elements, but rather the *relative number of times* each manifold is represented.

2. Parameterization of $E^{\mathcal{W}}_{\vec{\alpha}}$

We see in Fig. 3 how an observable may be used to parameterize the ensemble $E^{\mathcal{W}}_{\vec{\alpha}}$. For any given ensemble, there will in general be more than one observable X that we can use to parameterize the ensemble. Each observable is associated with a probability distribution function, which we may interpret as a mapping into the observable-parameterized ensemble $E^{\mathcal{W}}_{\vec{\alpha}}$ that gives us the shading depicted in the figure. Of course, this idea only makes sense if there is some sort of reference parameterization of the ensemble, from which we can do the mapping. In our visualization scheme of our figure, the depiction of our ensemble in its reference parameterization will have a constant shading. Later in this paper, we will discuss further the concept of reference parameter.

3. Collapse always proceeds from an ensemble to a subensemble

Consider the tree diagram of Fig. 4 A. This represents a typical experimental setup, with $\mathcal{O}_o$ representing the physical state of the observer at the beginning of an experiment, and the $\mathcal{O}_{\vec{\alpha}}$ representing three distinct future states. The top panel shows the corresponding ensembles whose boundary conditions are the *three*-object $\mathcal{O}_o$, or the *four*-objects $\mathcal{Q}_{\vec{\beta}'}$, $\mathcal{Q}_{\vec{\beta}''}$, or $\mathcal{Q}_{\vec{\beta}'''}$ Let us assume that the four-objects $\mathcal{Q}_{\vec{\beta}'}$, $\mathcal{Q}_{\vec{\beta}''}$, and $\mathcal{Q}_{\vec{\beta}'''}$ represent the *exhaustive* set of four-objects of length T that take their origin at $\mathcal{O}_o$. Since each of the four-objects $\mathcal{Q}_{\vec{\beta}}$ contain the three-object $\mathcal{O}_o$, then any world manifold $\mathcal{W}$ that contains any of the four-objects $\mathcal{Q}_{\vec{\beta}}$ must also contain the three-object $\mathcal{O}_o$. Therefore, each of the ensembles $E^{\mathcal{W}}_{\vec{\beta}}$ must be subsets of the ensemble $E^{\mathcal{W}}_o$, that is, $E^{\mathcal{W}}_{\vec{\beta}} \subseteq E^{\mathcal{W}}_o$. Since the four-objects are exhaustive, we have $E^{\mathcal{W}}_o = E^{\mathcal{W}}_{\vec{\beta}'} \cup E^{\mathcal{W}}_{\vec{\beta}''} \cup E^{\mathcal{W}}_{\vec{\beta}'''}$, as depicted in the upper panel. (Said differently: the $E^{\mathcal{W}}_{\vec{\beta}}$ cover $E^{\mathcal{W}}_o$.) As discussed previously, speaking in loose terms, we might describe the progression from $\mathcal{O}_o$ to one of the $\mathcal{O}_{\vec{\alpha}}$ as a "collapse" from the ensemble $E^{\mathcal{W}}_o$ onto one of the three subensembles $E^{\mathcal{W}}_{\vec{\beta}}$.

Recall one of our earlier premises: The state of a physical object $\mathcal{O}$, and the world ensemble relative to this object, is a function solely of the three-object $\mathcal{O}$ itself, and not of any particular four-object that contains $\mathcal{O}$. (See Fig. **??** A.) A corollary to this notion is that the world ensemble relative to an observer should *not* in any way require any "memory" of "where the object has been in the past." Therefore, it would be preferable if we could express the above relationship in terms of ensembles that are bounded by three-objects only. That is, we would like to say that $E^{\mathcal{W}}_o = E^{\mathcal{W}}_{\vec{\alpha}'} \cup E^{\mathcal{W}}_{\vec{\alpha}''} \cup E^{\mathcal{W}}_{\vec{\alpha}'''}$. Another way of stating this equation would be that the progression from $\mathcal{O}_o$ to one of the $\mathcal{O}_{\vec{\alpha}}$ could be described, speaking loosely, as a "collapse" from the ensemble $E^{\mathcal{W}}_o$ onto one of the three subensembles $E^{\mathcal{W}}_{\vec{\alpha}}$. Thus, we would like to assume that each of the "downstream" ensembles $E^{\mathcal{W}}_{\vec{\alpha}}$ is a subset of the "upstream" ensemble, $E^{\mathcal{W}}_o$.

4. The $E^{\mathcal{W}}_{\vec{\alpha}}$ are fractal

Through a symmetry argument, however, we soon see that the above consideration is problematic. The root of the problem is that it is not clear how to assign in rigorous terms which of two connected three-objects is "downstream" of the other. Consider the diagram of Fig. 4 B, in which we have two three-objects $\mathcal{O}_{\vec{\alpha}'}$ and $\mathcal{O}_{\vec{\alpha}''}$ with temporal supports $\vec{t}'$ and $\vec{t}''$. We assume that there exists at least one four-object manifold $\mathcal{Q}$ that joins them. Each of these three-object manifolds maps to an ensemble $E^{\mathcal{W}}_{\vec{\alpha}'}$ and $E^{\mathcal{W}}_{\vec{\alpha}''}$. What is the relationship between $E^{\mathcal{W}}_{\vec{\alpha}'}$ and $E^{\mathcal{W}}_{\vec{\alpha}''}$? We know that these two ensembles are not completely disjoint, since any world manifold $\mathcal{W}$ that contains $\mathcal{Q}$ must also contain $\mathcal{O}_{\vec{\alpha}'}$ as well as $\mathcal{O}_{\vec{\alpha}''}$. Thus we may entertain the notion that $E^{\mathcal{W}}_{\vec{\alpha}'} = E^{\mathcal{W}}_{\vec{\alpha}''}$. However, this will not be true in every case, so let us consider the more interesting scenario that $E^{\mathcal{W}}_{\vec{\alpha}'} \neq E^{\mathcal{W}}_{\vec{\alpha}''}$. Through a symmetry argument we may consider that a typical relationship would be that depicted in the upper panel of Fig. 4 B, in which $E^{\mathcal{W}}_{\vec{\alpha}'}$ and $E^{\mathcal{W}}_{\vec{\alpha}''}$ overlap, but neither is a subset of the other. This symmetry argument is compelling, since we would prefer to retain the generality of our assumption that the elements of $E^{\mathcal{W}}_{FUND}$ do not come with a predefined identification of the time variable. (Indeed, we do not even assume that the $\mathcal{W}$ must be time-orientable.) This implies that *any* choice of temporal support should be considered a valid one. So, if $\vec{t}$ is the temporal support of $\mathcal{Q}$ such that $\mathcal{O}_{\vec{\alpha}'}$ is "upstream" of $\mathcal{O}_{\vec{\alpha}''}$, then $-\vec{t}$ is the temporal support of $\mathcal{Q}$ such that $\mathcal{O}_{\vec{\alpha}'}$ is "downstream" of $\mathcal{O}_{\vec{\alpha}''}$.

We therefore arrive at a *seeming* contradiction: if we wish to retain the notion that "collapse" always proceeds from an ensemble $E^{\mathcal{W}}_1$ onto a subensemble $E^{\mathcal{W}}_2$ where $E^{\mathcal{W}}_2$ is always a subset of $E^{\mathcal{W}}_1$, then we must give up the symmetry depicted in Fig. 4 B, or else we will have simultaneously $E^{\mathcal{W}}_{\vec{\alpha}'} \subset E^{\mathcal{W}}_{\vec{\alpha}''}$ and $E^{\mathcal{W}}_{\vec{\alpha}''} \subset E^{\mathcal{W}}_{\vec{\alpha}'}$, which implies that an ensemble can be a subset of itself: $E^{\mathcal{W}}_{\vec{\alpha}'} \subset E^{\mathcal{W}}_{\vec{\alpha}'}$.

We quickly recognize, however, that the notion that a set is a subset of itself is not impossible; rather, this merely suggests that the $E^{\mathcal{W}}_{\vec{\alpha}}$ are self-similar, i.e., they are *fractal*. The fractal nature of $E^{\mathcal{W}}_{\vec{\alpha}}$ is illustrated in Fig. 4 C, in which we have a three-object $\mathcal{O}_{\vec{\alpha}'}$ that loops back upon itself ([31]). Thus, $\mathcal{O}_{\vec{\alpha}'}$ is "downstream" of itself so that $E^{\mathcal{W}}_{\vec{\alpha}'}$ is a subset of itself: $E^{\mathcal{W}}_{\vec{\alpha}'} \subset E^{\mathcal{W}}_{\vec{\alpha}'}$, as illustrated in the upper panel of Fig. 4 C. A similar situation arises in fractals such as the Mandelbrot set, illustrated in Fig. 5 A, in which there are (approximately?) one-to-one mappings between the entire image and each of the two rectangular boxes drawn in the image. Returning to Fig. 4 C, we have an infinitely recursive relationship: $\ldots \subset E^{\mathcal{W}}_{\vec{\alpha}'} \subset E^{\mathcal{W}}_{\vec{\alpha}'} \subset E^{\mathcal{W}}_{\vec{\alpha}'} \subset E^{\mathcal{W}}_{\vec{\alpha}'} \subset \ldots$. We can imagine that each traversal of the closed timelike curve in state space along the forward direction of proper time represents a "collapse" from $E^{\mathcal{W}}_{\vec{\alpha}'}$ onto the "next lowest level."

5. The direction of time

A corollary to this notion is the idea that following a trajectory "backwards in time" (more appropriately stated: replacing $\vec{t}$ with $-\vec{t}$) would result in going from $E^{\mathcal{W}}_{\vec{\alpha}'}$ to the next *highest* level. In more general terms, going forward in time along a trajectory (in the direction of $\vec{t}$) will always correspond to a transition from an ensemble onto a subensemble. Conversely, if we follow a trajectory "backwards" in time (in the $-\vec{t}$ direction), then all transitions will be from an ensemble to a covering ensemble. Essentially, this is how we *define* forwards versus backwards alond the $\vec{t}$ axis. Effectively, we have arrived at some insight into the problem of the "direction of time" – a difficult topic that deserves attention, although we will not go into it further in the present work. (Might this have some relation to CPT violation?) Time symmetry: with respect to an individual 4-manifold, the RHF is time-symmetric. But with respect to an ensemble, the RHF is *not* time symmetric, as we have just seen.

6. Each manifold is represented an infinite number of times in $E^{\mathcal{W}}_{\vec{\alpha}}$

Given the fractal characteristic of ensembles as depicted in Fig. 4 C, we see that we must accept the notion that any given individual manifold $\mathcal{W}$ may be present more than once within an ensemble $E^{\mathcal{W}}_{\vec{\alpha}}$. Given any individual 3-object $\mathcal{O}_{\vec{\alpha}}$, we propose that the number of "copies" of any given manifold $\mathcal{W}_{\vec{\gamma}}$ in $E^{\mathcal{W}}_{\vec{\alpha}}$ is equal to the number of occurrences of the three-object $\mathcal{O}_{\vec{\alpha}}$ in $\mathcal{W}_{\vec{\gamma}}$.

7. Fractal dimension of ensembles and topology of manifolds

A thorough investigation of the fractal characteristics of $E^{\mathcal{W}}_{\vec{\alpha}}$ – such as the various fractal dimensions (e.g. its correlation, information, and/or capacity dimensions) is beyond the scope of the present work. However, we will make a brief speculation regarding fractal dimension. Consider the two closed timelike curves (loops) depicted in Fig. 4 C. We might imagine that traversal of the inner CTC results in "collapse" to the next lower level, whereas traversal of the outer CTC results in collapse, say, two levels down. Let us call this number the "collapse number" of the loop. We offer the speculation that the collapse number of a loop may correspond in some fashion to some topological property of the loop. For example, consider the equivalence class of the loop under the equivalence relation of homotopy. This class is characterized by a number of invariants, such as "power," "conjugacy class," and "twist." We might imagine that the collapse number of the loop is a function of one or more of these topological invariants. As we will see in Sec. **??**, we would like to relate this topological invariant to the integral of a function along the loop. The investigations of such relations are naturally the purview of the study of the fundamental group and the study of Morse theory, which forms the basis of string theories. Future work, therefore, holds the promise of developing mathematical connections between the RHF and string theory.

8. *The probability measure of an ensemble*

We would like to define a "probability measure" $m\left[E^{\mathcal{W}}\right]$ of an ensemble $E^{\mathcal{W}}_{\vec{\alpha}}$ that corresponds to the probability that an object $\mathcal{O}_o$ will evolve to the state $\mathcal{O}_{\vec{\alpha}}$. In slang terminology, we desire $m\left[E^{\mathcal{W}}\right]$ to correspond to the probability of "collapse" from $E^{\mathcal{W}}_o$ onto a subset $E^{\mathcal{W}}_{\vec{\alpha}}$, according to the equation:

$$P = m\left[E^{\mathcal{W}}_{\vec{\alpha}}\right]/m\left[E^{\mathcal{W}}_o\right] \tag{1}$$

For the RHF to live up to the claim that it abides by classical notions of probability, we would like for our probability measure m to be a real number that is always nonnegative. Furthermore, if we normalize the probability measure of the ensemble that corresponds to the observer at the beginning of an experiment to one, $m\left[E^{\mathcal{W}}_o\right] = 1$, then we would like for the measure of any subensemble of $E^{\mathcal{W}}_o$ to take values between 0 and 1. Finally, we desire that our probability measures satisfy what we will refer to as a "linearity" criterion: If $E^{\mathcal{W}}_o$ is the union of disjoint ensembles $E^{\mathcal{W}}_i$, then

$$m\left[E^{\mathcal{W}}_o\right] = \sum_i m\left[E^{\mathcal{W}}_i\right] \tag{2}$$

In other words, if $E^{\mathcal{W}}_1$ and $E^{\mathcal{W}}_2$ are disjoint subsets of $E^{\mathcal{W}}_o$, then the probability of collapse onto $E^{\mathcal{W}}_1$ *or* $E^{\mathcal{W}}_2$ equals the probability of collapse onto $E^{\mathcal{W}}_1$ *plus* the probability of collapse onto $E^{\mathcal{W}}_2$.

We see quickly that our above criteria will be satisfied if we conceptualize the probability measure of an ensemble as something akin to the area or volume of the given ensemble. In other words, we think of an ensemble as a picture or an image, as depicted in Fig. 5 A, with the probability measure of the ensemble or any given subensemble as the area of that portion of the image. (Even though we have depicted $E^{\mathcal{W}}$ in Fig. 5 A as being a two-dimensional space, it is trivial to make the basic notion of probability measure-as-area to work regardless of the dimensionality of the space of world manifolds $E^{\mathcal{W}}$. If $E^{\mathcal{W}}$ is three-dimensional, then we think of measure as volume instead of area; and so on. At this point, we have not figured out the dimensionality of the space of world manifolds $E^{\mathcal{W}}$.)

In simplistic terms, we can think of the measure of a given ensemble as being proportional to the "number of manifolds $\mathcal{W}$" that are in the ensemble, in much the same way that we think of the area of an image as being proportional the "number of points" in the image, or the length of the line segment $[a, b]$ along the real axis as being the "number of points" between a and b. Strictly speaking, of course, this is incorrect, as we realize when we consider that for any two line segments $[a, b]$ and $[c, d]$ there is a one-to-one mapping between them; therefore they have the same "number of points" even if they are of differing lengths. To continue the analogy, the mapping in Fig 3 A from $E^{\mathcal{W}}$ in its "natural paramaterization" to $E^{\mathcal{W}}$ parameterized by the observable X is one-to-one. We will assume that the measure of the ensemble $E^{\mathcal{W}}$ will correspond to the volume of the ensemble as parameterized in its natural

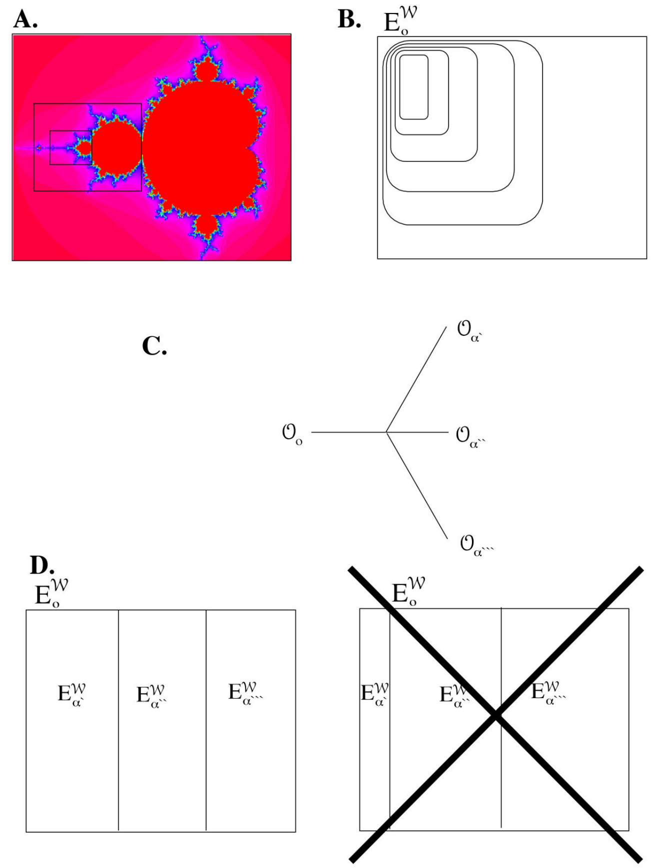

FIG. 5: **A.** The Mandelbrot set. The black boxes illustrate the image's self-similarity. **B.** An ensemble, with black boxes that illustrate self-similarity. **C.** A trajectory tree with a vertex with $N = 3$. **D.** Two potential ensemble diagrams to correspond to the trajectory tree in C. We postulate that the area of any given ensemble equals the probability measure of collapse onto that ensemble. According to the random walk hypothesis, each branch of the trajectory must be equally likely. Therefore, the ensemble diagram on the right is inconsistent with the random walk hypothesis.

dimension, as opposed to parameterized in some other way (e.g., using some observable X).

By analogy to the image of the Mandelbrot set, we can see that when we speak about the "area" of a portion of the image, e.g. the area of the black box in Fig. 5 A, it is necessary to provide some information regarding the scale. That is, the two small boxes in Fig. 5 A represent the same set, but different areas. This is because it is the same set but at a different scale. Note here that the notion of scale is not an absolute variable, but is instead a relative value. (Like voltage.) Since $E^{\mathcal{W}}$ is fractal, we see that the measure of an ensemble $m\left[E^{\mathcal{W}}\right]$ must likewise make reference to scale; e.g., in Fig. 4 C, before we can specify the measure of the ensemble $E^{\mathcal{W}}_{\vec{\alpha}'}$, we must specify *which* of the boxes in the upper panel of Fig. 4 C we are referring to – i.e., we must specify the scale.

In the current paper, we are not going to go any further into how exactly we define scale. Letu us simply assert that there exists some scale factor s, and that whenever we refer to the measure of an ensemble $E^{\mathcal{W}}$, we must specify the scale factor s as well as the ensemble itself; together, we will denote the measure of an ensemble E at scale s as $m\left[E^{\mathcal{W}}(s)\right]$. And if $E_2^{\mathcal{W}}(s_2)$ is a subset of $E_1^{\mathcal{W}}(s_1)$, then we can calculate the probability of collapse from $E_1^{\mathcal{W}}(s_1)$ onto $E_2^{\mathcal{W}}(s_2)$ as the measure of $E_2^{\mathcal{W}}(s_2)$ divided by the measure of $E_1^{\mathcal{W}}(s_1)$:

$$P\left[E_1^{\mathcal{W}} \rightarrow E_2^{\mathcal{W}}\right] = m\left[E_2^{\mathcal{W}}(s_2)\right]/m\left[E_1^{\mathcal{W}}(s_1)\right] \quad (3)$$

In more general terms, we will define a decoherence functional $d\left[E_1^{\mathcal{W}}(s_1), E_2^{\mathcal{W}}(s_2)\right]$ to be the probability of collapse from $E_1^{\mathcal{W}}(s_1)$ onto $E_2^{\mathcal{W}}(s_2)$:

$$P\left[E_1^{\mathcal{W}} \rightarrow E_2^{\mathcal{W}}\right] = d\left[E_1^{\mathcal{W}}(s_1), E_2^{\mathcal{W}}(s_2)\right] \quad (4)$$

Of course, we can only make sense of this if $E_2^{\mathcal{W}}(s_2)$ is a subset of $E_1^{\mathcal{W}}(s_1)$. We would like to define the decoherence functional in more general terms so that it may take as input any two arbitrary ensembles $E_1^{\mathcal{W}}(s_1)$ and $E_2^{\mathcal{W}}(s_2)$. The reason we wish to do this is so that we can draw comparison between the formalism of the RHF and the formalism of the consistent histories formulation (CHF), which we will discuss below. Indeed, one of the central concepts of the CHF is the decoherence functional, which takes two histories as input and produces a complex number. In loose terms, the decoherence functional of the CHF plays a role analogous to the role of the projection operator in the standard formulation of quantum mechanics. An important feature of any decoherence functional is that it is *Hermetian*. Therefore, we will make the assumption that the decoherence functional of the RHF must also be Hermetian:

$$d\left[E_1^{\mathcal{W}}(s_1), E_2^{\mathcal{W}}(s_2)\right] = d^*\left[E_2^{\mathcal{W}}(s_2), E_1^{\mathcal{W}}(s_1)\right] \quad (5)$$

(where the asterisk denotes complex conjugate).

In the CHF, in general, there may be more than one functional that satisfies the requirements of being a decoherence functional. So for the sake of illustration, let us propose a functional that meets all of the requirements that we have come up with so far to be a decoherence functional for the RHF. Suppose we are given two completely arbitrary ensembles $E_1^{\mathcal{W}}(s_1)$ and $E_2^{\mathcal{W}}(s_2)$. Since we are speaking in completely general terms, we make no assumption that either one is or is not a subset of the other; they may partially overlap, as in Fig. 4 B. We will propose that the decoherence functional is a complex number $d\left[E_1^{\mathcal{W}}(s_1), E_2^{\mathcal{W}}(s_2)\right] = d_R + id_C$ with d_R and d_C real-valued nonnegative numbers, where d_R is the measure of the intersection of the two ensembles divided by the measure of the union of the two ensembles, and d_C is the product of three numbers: the measure of $E_1^{\mathcal{W}}(s_1)$ minus the measure of the intersection; the measure of $E_2^{\mathcal{W}}(s_2)$ minus the measure of the intersection; and the measure of $E_1^{\mathcal{W}}(s_1)$ minus the measure of $E_2^{\mathcal{W}}(s_2)$. We propose this as one possible decoherence functional (as is the case in the CHF, there may be more than one decoherence functional) because it satisfies the requirement of Hermiticity, and also satisfies the requirement that if $E_2^{\mathcal{W}}(s_2)$ is a subset of $E_1^{\mathcal{W}}(s_1)$ – in which case, d_C is zero – then $d\left[E_1^{\mathcal{W}}(s_1), E_2^{\mathcal{W}}(s_2)\right]$ is the probability of collapse from $E_1^{\mathcal{W}}(s_1)$ to $E_2^{\mathcal{W}}(s_2)$.

E. The process of collapse

Is collapse a continuous process? Or a discrete event? A discrete event would be like jumping from an ensemble onto a subensemble, as in Fig. 5 C and D. A continuous process would be like slowly and continuously "increasing (or decreasing) the magnification" so that we "zoom in (or out)" of the images of Fig. 5 A and B. From that point of view, as we progress forward down the branches of the tree in a tree diagram, we slowly and continuously zoom in on the image. The difference between any two trajectories, therefore, might be: which edge of the image is being "shaved off" as we zoom inwards?

For now, we leave this question (whether collapse is a continuous or a discrete event) an open one. Intuitively, we favor the notion that it is a continuous process in rigorous terms, but one that can be modelled as a discrete process under appropriate approzimation techniques. According to this notion, the tree diagram of Fig. 5 C – which models collapse onto one of three distinct final states as a discrete process, with each of the three final states being realized with probability 1/3 – would be considered to be an approximation of the "true" tree diagram. We would assume that for the approximation tree diagram to be valid, the "true" diagram must give rise to the same probability distribution. Looking at the ensemble (on the left) of Fig. 5 D, the discrete approximation would correspond to a discrete jump from $E_o^{\mathcal{W}}$ onto one of the subensembles $E_{\vec{\alpha}}^{\mathcal{W}}$; we would implicitly understand that the "exact" transition from $E_o^{\mathcal{W}}$ onto one of the three subensembles would be analogous to slowly "zooming in" from $E_o^{\mathcal{W}}$ onto one of the $E_{\vec{\alpha}}^{\mathcal{W}}$. Regardless of whether we use the discrete approximation technique or the exact technique, however, the measures of each of the subensembles $E_{\vec{\alpha}}^{\mathcal{W}}$ could be used to calculate the probability of collapse onto that subensemble.

F. UE is a Hilbert space

The $E_{\vec{\alpha}}^{\mathcal{W}}$ in the RHF play a role analogous to that played by the wavefunction ψ in the Schrödinger formulation of QM. One of the central features of the Schrödinger formalism is that the ψ define a Hilbert space. Therefore, it is only natural to conceive of the $E_{\vec{\alpha}}^{\mathcal{W}}$ as likewise defining a Hilbert space. To accomplish this goal, we must define the operations of addition, scalar multiplication, and inner product. Let us see if we can do this explicitly.

We begin with specification of the elements of our Hilbert space as *any* ensemble $E_i^{\mathcal{W}}$ of world manifolds. The set of all possible $E_i^{\mathcal{W}}$ will be denoted $\mathcal{UE}$ (universal world ensemble space). $E_{FUND}^{\mathcal{W}}$ and the $E_{\vec{\alpha}}^{\mathcal{W}}$ are specific types of elements of $\mathcal{UE}$. In other words, it is possible to define an ensemble $E_1^{\mathcal{W}}$ that *cannot* be defined in terms of a single $\vec{\alpha}$; for example, an ensemble that has a finite number of individual world manifolds. In general, the notation $E_i^{\mathcal{W}}$, where i is some digit, will be used to refer to an element of $\mathcal{UE}$ if we do not wish to specify whether it takes the specific form: $E_{\vec{\alpha}}^{\mathcal{W}}$.

Addition of two ensembles $E_1^{\mathcal{W}}$ and $E_2^{\mathcal{W}}$ is naturally defined as their union: $E_1^{\mathcal{W}} \cup E_2^{\mathcal{W}}$. It is trivial to see that $\mathcal{UE}$ is closed under the union operation.

Scalar multiplication $E_1^{\mathcal{W}} = k * E_2^{\mathcal{W}}$ is defined such that $E_1^{\mathcal{W}}$ is the union of k copies of $E_2^{\mathcal{W}}$. Thus, if the world manifold $\mathcal{W}$ is represented n times in $E_2^{\mathcal{W}}$, then it will be represented $n * k$ times in $E_1^{\mathcal{W}}$.

We define an inner product $\langle E_1^{\mathcal{W}}, E_2^{\mathcal{W}} \rangle$ which takes two ensembles and yields a complex number. It must satisfy the axioms of linearity, symmetry, and positive definiteness. The metric of our inner product space is defined as the norm of the inner product of $E_1^{\mathcal{W}}$ with itself: $|E_1^{\mathcal{W}}| = \sqrt{\langle E_1^{\mathcal{W}}, E_1^{\mathcal{W}} \rangle}$. This allows us to define the notion of the "difference" between two ensembles as the norm of their difference: $|E_1^{\mathcal{W}} - E_2^{\mathcal{W}}|$.

G. Overview of the various spaces of the RHF: unanswered questions

For reference, Fig. 6 presents an overview of the various spaces that have been introduced in this section. Note that there is a (one-to-one?) mapping between the elements of $\mathcal{UE}$, the elements of $\mathcal{O}$-state space, and the elements of the Fundamental $\mathcal{O}$-ensemble.

We should review at this point some characteristics of these spaces that have *not* been addressed. What are the topologies of the $\mathcal{W}$-, $\mathcal{Q}$-, and $\mathcal{O}$-state spaces? Are they open, closed, simply connected, multiply connected?

1. *$\mathcal{O}$-State Space: Connection to string theory?*

According to the above considerations, the state of $\mathcal{O}_o$ can be fully specified simply by providing $g_{ab}(0,0,0,0)$ and its derivatives (as well as its temporal support $\vec{t}$). This makes $\vec{\alpha}$ an infinite dimensional vector, since the number of derivatives of g_{ab} is infinite.

To show this explicitly, let us write out explicitly the components of $\vec{\alpha}$. g_{ab} is a 4×4 matrix with sixteen components. However, the metric tensor g_{ab} is, by definition, symmetric (that is, $g_{ab} = g_{ba}$) and therefore actually requires only ten components rather than sixteen for its complete specification. Denote these ten components using the notation $g_j(p), j = 1, 2, 3, \ldots, 10$. To perform a Taylor series expansion about the point p, we must specify the value of each of these ten components as well as all combinations of partial derivatives. That is, we must specify: $\nabla^i g_j(p), j = 1, 2, \ldots, 10; i = 0, 1, 2, \ldots, \infty$

Now let us consider a ten dimensional space defined by the ten components of $g_j(x_1, x_2, x_3, x_4)$. Consider a curve of arbitrarily small length S, parameterized by the parameter s: $g_j(x_1(s), x_2(s), x_3(s), x_4(s)) = g_j(s)$. Theorem: If we are given the value of $g_j(s)$ for $s \in [s_i, s_f]$ – that is, we are given an arbitrarily short line segment in the ten dimensional space g_j – then we can

2. *Dimensions of space-time versus the dimensions of state space*

Given that our world manifolds $\mathcal{W}$ have four spatial dimensions with curvature described by symmetric metric g_{ab} that is 10 dimensional, our state space is (as we have just seen) 10 dimensional. If our physical world were to have a different number of spatial dimensions, then how many dimensions would we have for state space? Answer: if we were to live in a universe of 2, 3, 4, 5, 6, ... spatial (spacetime) dimensions, then using the logic of our preceeding discussion, our state space would have 3, 6, 10, 15, 21, ... dimensions, respectively.

III. DISCUSSION

APPENDIX A: TAYLOR'S THEOREM

Taylor's Theorem is summarized from Ref. [6], sec. 45. Suppose that a function f is analytic throughout an open disk $z - z_0 < R_0$, centered at z_0 and with radius R_0. Then, at each point z in that disk, $f(z)$ has the series representation

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n \tag{A1}$$

where

$$a_n = \frac{f^{(n)}(z_0)}{n!} \tag{A2}$$

Taylor's Theorem can be easily generalized to a real function of n variables [31]. For the purposes of Sec. II B, we make use of such a generalization of Taylor's Theorem. Instead of a single-valued function f, we have function g_{ab} with sixteen components: $g_{11}, g_{12}, g_{13}, \ldots, g_{44}$, further reduced to ten components since we assume that g_{ab} is symmetric. In place of the single variable z, we have four variables (x_1, x_2, x_3, x_4).

Let us see how this allows us to express $\vec{\alpha}$ as an infinite dimensional vector. Denote the ten components of g_{ab} in a fashion to make explicit the fact that it has ten components: $g_1, g_2, g_3, \ldots, g_{10}$. (That is, $g_j, j = 1, 2, \ldots, 10$.) To perform a Taylor series expansion about the point p, we must specify the value of each of these ten components as well as all combinations of partial derivatives. That is, we must also specify: $Del^i g_j(p), j = 1, 2, \ldots, 10; i = 0, 1, 2, \ldots, \infty$

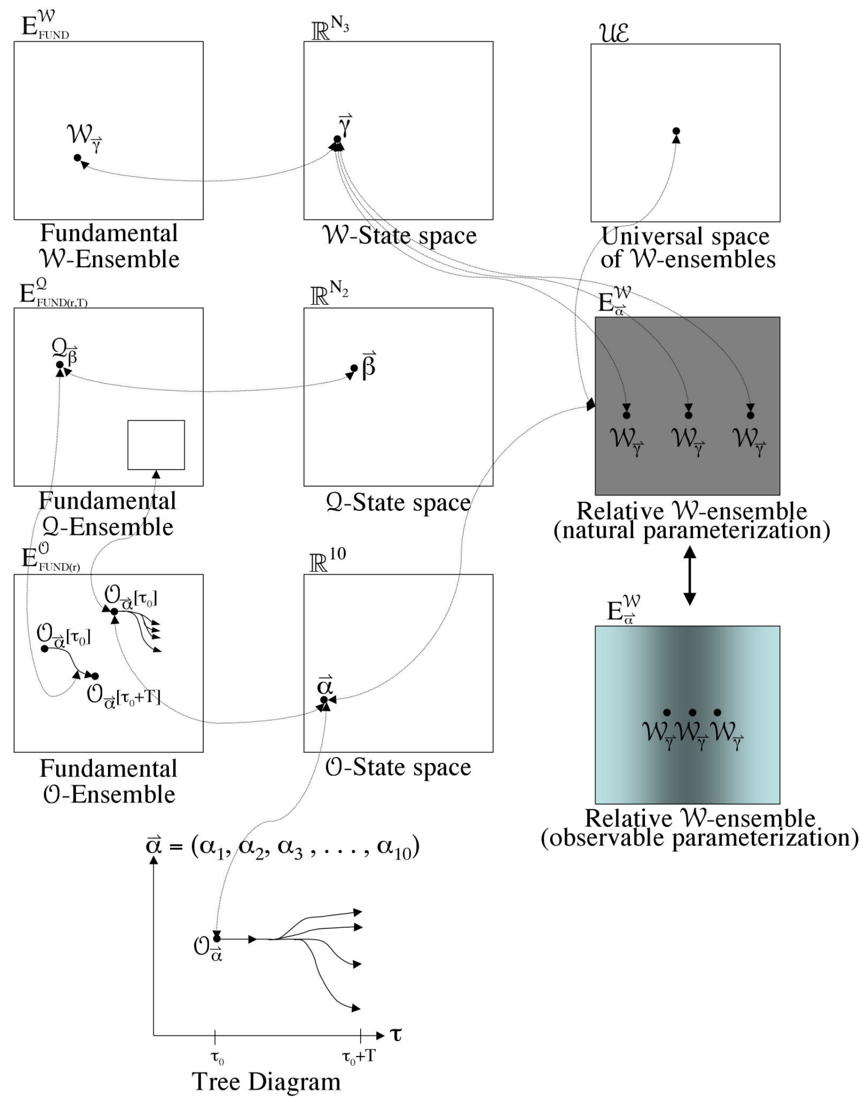

FIG. 6: Overview of the various spaces that are used in the RHF. The fundamental $\mathcal{W}$-ensemble is an ensemble of world manifolds. Each element in this space is unique – in fact, each element has a unique topology. A "relative $\mathcal{W}$-ensemble" is defined in terms of the object manifold $\mathcal{O}_{\vec{\alpha}}$. In this case, any given element $\mathcal{W}$ will typically be represented more than once. State space is ten dimensional. Don't know whether state space is simply or multiply connected, flat or curved. Same goes for the universal world ensemble space. The RHF is constructed from a single state space and a single universal world space. There are lots of history ensembles – in fact, one for each element of state space. $\mathcal{UE}$ is a Hilbert space.

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